

The Catalyst

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Goal Setting

Start to ask yourself:

- ✓ **What do I want? :**
 - What am I passionate about? What do I want to become an expert in?
 - What are my long-term goals, both personally and professionally?
- ✓ **How do I get it?**
 - What is my plan to attain those long-term goals?
 - What manageable short-term goals can I put in place now to help me achieve them?
- ✓ **Who/What do I need to help me get it?**
 - Who in my network can mentor, assist, and/or advocate for me?
 - What degrees, skillsets, and/or certifications will I need?
- ✓ **How long will it take me to get it?**
 - How can I develop a realistic timeline to completion of my plan? (Begin with the end in mind!)

Professional Mission Statement:

Long-Term Professional Goals:

1. *Become a tenured Professor with a group focused on developing next-generation prosthetics using dynamic 3-D mathematical modeling.*

Short-Term Goals:

- a) *Get a SM/PhD in Biomedical Engineering.*
- b) *Get internship experience at a prosthetics startup.*
- c) _____

2. _____
- _____

Short-Term Goals:

a) _____

b) _____

c) _____

3. _____

Short-Term Goals:

a) _____

b) _____

c) _____

Long-Term *Personal* Goals:

1. _____

Short-Term Goals:

a) _____

b) _____

c) _____

2. _____

Short-Term Goals:

a) _____

b) _____

c) _____

3. _____

Short-Term Goals:

a) _____

b) _____

c) _____

Part I: The Road Less Traveled

Preparation for Graduate School

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Summary of Part I Slides

Slide 1

Research is your fuel

Summer research programs:

- @ Research I institutions
- @ Your home institution
- @ National labs

Summer internships:

- @ Corporations (Fortune 500 v. startups)
- @ National labs
- @ Government agencies (federal v. state)
- @ Non-profit agencies

Notes:

Slide 2-5

Applying for Summer Research

- How to **find**?
 - Refer to the Catalyst!
 - Start close to home! (search: home/nearby institutions)
 - Use your network (ask: profs/family/friends)
 - Campus career office/program admins
 - On-campus/local research conferences
- How to **apply**?
 - “mini-grad school application process”
 - Online application
 - Transcript
 - Recommendations
 - Essays
 - Supplemental information (skills, preferences, etc.)
- How to be **competitive**?
 - GPA
 - Unique skill sets (programming skills/lab experience)
 - Personalized resume for each program
 - Mission statements/clear goals
 - Great recommendations

Notes:

Slide 6

Coursework will serve as warning signs

- Increase your GPA as you increase coursework in your major
- Focus on doing well in your major and upper level courses
- Challenge yourself
- Undergraduate coursework is the **foundation** for grad school

Notes:

Slide 7

Building relationships will provide you direction and guidance

- Advisors: give advice on coursework/internships/research direction
- Counselors: formally guide your academic and career choices
- Mentors: (professional/personal/peers) give informal advice and support
- Advocates: speak on your behalf on the application

Notes:

Slide 8

Pursue your passions: "Pleasure in the job puts perfection in the work."

- Aristotle

Notes:

Yearly Planner

Freshman Year

Potential Major(s): _____

Potential Minor(s): _____

GPA Goals:

Cumulative GPA:

Fall ____ / ____ Spring ____ / ____

Major GPA:

Fall ____ / ____ Spring ____ / ____

Organizations to join:

1) _____

2) _____

3) _____

Professional Development Goals:

1) _____

2) _____

3) _____

Community Service Opportunities:

1) _____

2) _____

3) _____

Community Service Goals:

1) _____

2) _____

3) _____

	Credits	Course	Core or Elective?	Grade
Fall				
Spring				

Potential Summer Internships/Research Opportunities: _____

Sophomore Year**Personal Goals:**

- 1) _____
- _____
- 2) _____
- _____
- 3) _____
- _____

GPA Goals:**Cumulative GPA:**

Fall ____ / ____ Spring ____ / ____

Major GPA:

Fall ____ / ____ Spring ____ / ____

Potential Semester Research:

- 1) _____
- _____
- 2) _____
- _____
- 3) _____
- _____

Professional Development Goals:

- 1) _____
- _____
- 2) _____
- _____
- 3) _____
- _____

**Potential Technical Conferences/
Educational Travel Opps:**

- 1) _____
- _____
- 2) _____
- _____
- 3) _____
- _____

Community Service Goals:

- 1) _____
- _____
- 2) _____
- _____
- 3) _____
- _____

	Credits	Course	Core or Elective?	Grade
Fall				
Spring				

Potential Summer Internships/Research Opportunities: _____

Junior Year**Personal Goals:**

- 1) _____

- 2) _____

- 3) _____

GPA Goals:**Cumulative GPA:**

Fall ____ / ____ Spring ____ / ____

Major GPA:

Fall ____ / ____ Spring ____ / ____

GRE Goals:

Quantitative ____ Verbal ____

Potential Semester Research:

- 1) _____

- 2) _____

- 3) _____

Professional Development & Service Goals:

- 1) _____

- 2) _____

- 3) _____

**Potential Technical
Conferences/ Educational
Travel Opportunities:**

- 1) _____

- 2) _____

**Potential Grad
Fellowships/Preparation:**

- 1) _____

- 2) _____

- 3) _____

**Graduate Schools to
Consider/Apply to Visit:**

- 1) _____

- 2) _____

- 3) _____

- 4) _____

- 5) _____

	Credits	Course	Core or Elective?	Grade
Fall				
Spring				

Potential Summer Internships/Research Opportunities: _____

Senior Year**Personal Goals:**

- 1) _____

- 2) _____

- 3) _____

GPA Goals:**Cumulative GPA:**

Fall ____ / ____ Spring ____ / ____

Major GPA:

Fall ____ / ____ Spring ____ / ____

GRE Goals:

Quantitative ____ Verbal ____

Potential Semester Research:

- 1) _____

- 2) _____

- 3) _____

Professional Development & Service Responsibilities:

- 1) _____

- 2) _____

- 3) _____

**Potential Technical
Conferences/ Educational
Travel Opportunities:**

- 1) _____

- 2) _____

Graduate Applications/ Fellowship Applications to Complete:

Graduate School App		Graduate Fellowships	

	Credits	Course	Core or Elective or Graduate Course*?	Grade
Fall				
Spring				

Potential Summer Internships/Research Opportunities: _____

Potential Recommenders Worksheet

Tips for Building Relationships with Potential Recommenders

- Purposefully develop a relationship with your **research supervisors**. Research recommendations are of the utmost importance. Recommenders need to be able to speak to your ability to succeed in a research environment. Set up regular research meetings to discuss progress, challenges, ideas, etc. Keep in touch after you leave via email.
- If you plan to ask a **professor** from a class for a recommendation, make sure they get to know you outside of the classroom. Ask to meet with them to discuss graduate school. Keep in touch with them to ensure they have continued access to your professional progress and quite possibly you will gain access to theirs (mutually beneficial prof/protégé relationship).
- Review the relationships you have with previous **internship managers, research supervisors** and **professors**. There may be existing recommenders ready for you to tap into.

Potential recommender 1: _____

Date of last correspondence: _____

Potential recommender 2: _____

Date of last correspondence: _____

Potential recommender 3: _____

Date of last correspondence: _____

Potential recommender 4: _____

Date of last correspondence: _____

Potential recommender 5: _____

Date of last correspondence: _____

Summer Research Programs

There are many opportunities available in the summer. Use this starter list to help you initiate your plans.

Colleges & Universities

- Albert Einstein College of Medicine Summer Undergraduate Research Program
- Amgen Scholars Program
- Berkeley SUPERB
- California Institute of Technology MURF
- Cornell LSAMP
- Duke University Summer Research Opportunity Program
- Georgia Tech SURE REU
- Harvard Summer Honors SHURP
- MIT Summer Research Program
- New York University MD/PhD SURP
- North Carolina State University REU+
- Northwestern Summer Research Program
- Pepperdine University – Seaver Undergraduate Research in Biology
- Rutgers REU
- Stanford Civil and Environmental Engineering Summer Programs
- Stanford Summer Research Program
- The Broad Institute Summer Research Program in Genomics
- University of California, Irvine SURF
- University of California, San Diego SRO
- University of Maryland, Baltimore County – BSURE
- University of Massachusetts Medical School Summer Research Fellowship Program
- University of Michigan at CERN
- University of Nebraska – Lincoln Summer Research Program
- University of North Carolina at Chapel Hill – SURE REU
- University of Oregon Summer Program for Undergraduate Research
- University of Pennsylvania Minority Health Internships
- University of Washington Undergraduate Research Program
- University of Wisconsin-Madison IBS-SRP
- Washington University Summer Research Programs

Government

- Center for Disease Control Internships
- HHMI Summer Experiences
- NASA NSTI
- National Cancer Institute
- National Institute of Allergy and Infectious Disease Summer Programs
- National Institutes of Health Undergraduate Scholarship Program
- National Science Foundation REU (Research Experience for Undergraduates)
- US Department of Energy

Other

- Pasteur Foundation Summer Programs
- UNCF MES (DOE)
- UNCF MUST
- Woods Hole

Coursework Strategies

1. Classes related to your major are most important. **Make sure you excel in these.**
2. A strong foundation of **core and relevant elective courses** is necessary to succeed in graduate school.
3. **Challenge yourself** by taking advanced, graduate level, courses in your area of interest.
4. **Layout your plan** for completion of coursework and update it every semester. Refer to your academic advisor to ensure that you are meeting all degree requirements.
5. Setup your course schedule in a way that **works best for when you work best**, then fill in your other critical responsibilities (i.e.- graduate applications, semester research, etc.) as if they were classes.

Part II: The Hitchhiker's Guide

Graduate Application Process

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Summary of Part II Slides

Slide 1

Be organized with respect to:

- Your undergraduate/graduate school/career goals
- Your application process
- Your network

Notes:

Slide 2

Start early

- In all aspects of the grad application process, being early is actually “on time” for many opportunities
- Prepare for and take the GRE early
- Apply for fellowships and admission early
- Prepare well in advance so that you go into the fellowship and admission application process knowledgeable about the institutions/departments/professors

Notes:

Slides 3 & 4

Be mindful that faculty review every application holistically

Recommendations

- Critical to have from someone you did research with
- Speaks to your ability to do research
- From faculty who know you well
- No character references from coaches, friends, or family

Research experience

- Key component – find a way to do research
- Beneficial for your own preparation and ability to contribute to the group you are joining
- Speaks to your ability to succeed in graduate school
- Opportunity to get to know faculty members

Statement of purpose/research

- Your chance to speak beyond the numbers
- Focused, why would you make a good researcher?

GPA/coursework

- Need to have a strong foundation in department classes
- Do well in core classes
- Need foundation to succeed in graduate school
- Challenge yourself with advanced courses

GRE/GRE subject tests/TOEFL and IELTS

- Quantitative score can be a red flag
- Take them early
- TOEFL or IELTS required for non-native speakers
- Retake if needed

Notes:

Graduate School Matrix

Institution	Programs of Interest	Professors of Interest	Deadlines	Application Requirements	Contact Information	Additional Notes

Institution	Programs of Interest	Professors of Interest	Deadlines	Application Requirements	Contact Information	Additional Notes

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Starter List of Graduate Fellowships

Federal

- American Institute of Physics State Department Fellowship
- Boren fellowships
- Center for Disease Control Ferguson Fellowship
- Department of Energy Computational Science
- Department of Energy National Nuclear Security Administration Science graduate fellowships
- National Defense Science and Engineering Graduate Fellowship
- National Institutes of Health PREDOC Fellowship
- National Science Foundation
- Smithsonian fellowships
- US Environmental Protection Agency National Network for Environmental Management Studies NNEMS student fellowships
- US Environmental Protection Agency Science to Achieve Results (STAR) fellowships

Private

- American Association of University Women Fellowships
- Association for Institutional Research Graduate Certificate/Fellowships
- Balfour Fellowship
- Ford Foundation Fellowship
- GEM Fellowship
- Harriet Jenkins PREDOC Fellowship
- Hertz Fellowship
- Hispanic Scholarship Fund
- Jack Anson Fellowship
- Jack Kent Cooke
- Jefferson Science Associates Fellowships
- Knowles Fellowships
- NASA Graduate Student Researchers Programs
- National Physical Science Consortium Fellowships
- National Urban Fellowship
- Rocky Mountain Fellowships
- Soros Fellowships for New Americans
- UNCF/MERCK Fellowships
- University of California Presidential PREDOC Fellowship
- University of New Mexico SUMMA Fellowship
- Woodrow Wilson Fellowships

And many others...

Calendar for Deadlines

Calendar Tips

- Separate out your tasks/deadlines by topics (use color-coding or your own system)
 - o Application-related
 - o Fellowship-related
 - o Academics
 - o Personal
 - o Etc.
- Give yourself more time than you think you need (at least 1 week before actual deadline)
- Identify “target months”
 - o **July/August:**
 - Draft personal statements
 - Draft shortlist of schools
 - Begin fellowship applications
 - Begin GRE preparation
 - Graduate Visit Programs deadlines
 - o **September:**
 - Send out statement of purpose for review
 - Register for GRE
 - Identify possible recommenders
 - Apply for graduate visit programs
 - o **October:**
 - Take GRE
 - Finalize statement of purpose
 - Begin submitting fellowship applications
 - Ask for recommendation letters
 - Order transcripts
 - o **November:**
 - Finalize all applications and recommendations
 - Confirm that all documents were received by the schools
- Don’t beat yourself up; adjust your weekly goals as necessary!

<u>Month:</u>							
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Weekly Goals

<u>Month:</u>							
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Weekly Goals

<u>Month:</u>							
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Weekly Goals

Application Tips

All parts of the application are important. Applications are reviewed by faculty members.

Recommendations

- 3 Minimum (at least 2 from faculty)
- Should speak to student's ability to do research –insight beyond the classroom!
- Recommender needs to know student well – seek out meetings/conversations with faculty.
- At least one should come from someone with whom you did research
- Faculty recommendations are best; strongest are from someone with an advanced degree
- Avoid character references – focus on your ability to do research and potential as a graduate student
- Recommender can include a "best student in X years, or student ranks in top X% of all the students I have had, etc."
- Give recommenders CV/resume and your statement of purpose

Research Experience

- Very important part of the application
- Make sure you participate in research: summer programs, undergraduate programs, industry program, etc. – seek opportunities inside and outside of your institution
- Make sure you build a relationship with your supervisor so he/she can write a strong letter of recommendation for you
- Consider summer research programs at top universities – *see sample list*
- Quality of research is important
- Look for research that is relevant to your interest area

Statement of Purpose

- Follow instructions – address all points, follow length limits
- Be clear, concise, and focused. This is not a personal/inspirational story
- Faculty read this – they care about specific examples that demonstrate you are a good researcher
- Helpful to identify a couple of faculty members whose research you are interested in. Do your homework, be knowledgeable
- Be focused, but not too narrow. Do not say there is only one specific thing or person you want to work with
- Why are you passionate about working in this field?
- Focus on your research interests and why you want to do research
- Have others review it: your advisor, trusted faculty mentors, advanced graduate students, etc.

GPA/Coursework

- Focus on core classes – math, science, engineering
- Challenge yourself with advanced classes in your discipline
- Have a plan for taking all the classes you need/want
- Low grades in major(core) classes are red flags
- Your coursework should show your interest in the research area
- GPA is just one piece of the application

GRE

- Quantitative is most important but do not ignore qualitative
- Low quantitative score is a red flag
- Non-native speakers need to take the TOFEL or IELTS
- Register early and take the test early

REMEMBER

- The application is a complete package. Reviewers look at trends from freshman to senior year, increases in intensity of coursework, consistency in research, major GPA, as a whole
- Upward trends are important, they show a steady improvement in performance, maturation of abilities and technical expertise
- Start your applications early!

Electronic Network

Who should you include?

- ☐ Professors
- ☐ Mentors/Advisors
- ☐ Peers who can provide guidance
- ☐ Deans
- ☐ Former employers

Why?

- ☐ Possible recommenders
- ☐ Possible reviewers for your application materials
- ☐ Professional/personal guidance
- ☐ Maintenance of meaningful connections

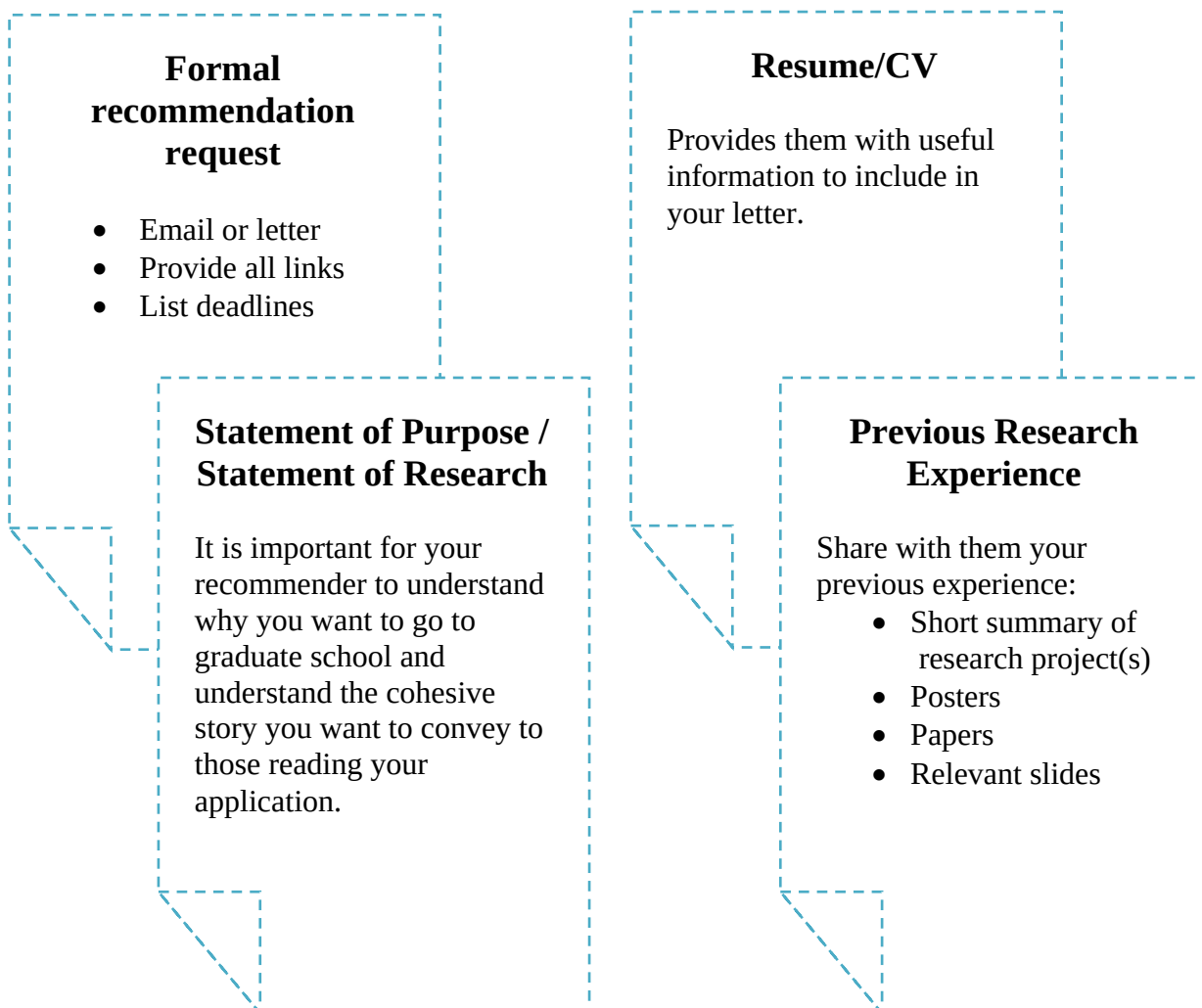
Affiliation	Contact Name	Email	Phone	Physical Address

Recommendation Letters

- Choose people who know you well and can speak to your abilities.
- Ask for recommendation letters in person and follow up with an email. The follow up email could be to thank them, provide more information, or to ask for a meeting to discuss further, depending on the initial interaction.
- Ask “Can you write a strong, positive recommendation?” If there is any doubt then you should pick someone else.
- Provide recommenders with a packet of all your information (below), if you want you can mention specific things you would like him or her to touch on. Ideally in person, but email is acceptable.

Recommender Packet

You should make a recommender packet for each person from whom you request a recommendation. Packets should include (but not limited to):



Statement of Purpose Guidelines

Things to know:

- o **Who** is reading this?
 - Faculty members – they are busy; be focused and concise
- o **What** are they interested in?
 - Students who can conduct research
 - Essentially hiring researchers who can deliver/make progress
 - Don't need to know your whole personal story – only relevant points
- o Read statement of purpose guidelines carefully. (Every program is different!)
- o **Avoid:**
 - CV/Resume in essay form
 - Grammatical/typographical errors – Have other people review your statement
 - Limiting interest to only one professor
 - Clichés

Points to include:

- ✓ Why do you want to go to graduate school?
- ✓ What specific area do you want to study/do research in?
- ✓ Why did you choose this research area? Why is it important?
- ✓ How does your previous research support your statement? (briefly explain your undergraduate research, what you learned from it, and how it led you to your current goals)
- ✓ Why do you want to study at this school, in this program/department? (Make it clear that you are familiar with this program/department)
- ✓ Why are you a **good fit** for this program?

Part III: Speed Dating

Selecting a School/Program/Advisor

Summary of Part III Slides

Slide 1

Selecting a **school**

1. Strategize: look at all your options holistically (research, culture, weather, city, community, etc). Plan ahead who you will contact, where you will visit, etc.
2. Visit: see what the school is actually like by going on campus, talking to the people, eating the food, facing the weather.
3. Ask: prepare a list of questions for students, professors and administrators.

Notes:

Slide 2

Selecting a **program**

1. Make sure you get the COMPLETE picture:
 - Potential advisors
 - Department policies: qualifying exams
 - Funding opportunities
 - Graduate student life
 - Housing
 - Length/focus/reputation of program
 - Teaching opportunities
 - Travel opportunities
 - Support mechanisms
 - Diversity landscape
2. What is important to you? Prioritize your wants vs. needs.

Notes:

Slide 3

Making an informed decision on funding: is graduate school free?

- Grants
- Fellowships
- Research Assistantships
- Teaching Assistantships
- Loans

Notes:

Slide 4

Selecting an **Advisor**

1. Prepare for the initial meeting (either on a grad visit or department preview weekend): know about their research, current publications, etc. Talk to their current grad students.
2. Ask knowledgeable questions about both their research and their advising. You are also interviewing them, in a sense.
3. Follow up! Send a thank you email, inquire about next steps.

Notes:

Starter list of Graduate Visit Programs

- Berkeley Edge
- CalTech Grad Preview
- Georgia Tech Focus
- Indiana University
- MIT Converge
- UC Davis Preview Day
- University of Florida CVD
- University of Maryland Preview Day
- University of Michigan PREVIEW
- University of Missouri Emerge
- University of Wisconsin-Madison OIE
- Virginia Tech Preview Weekend

Note: Many visit programs have summer application deadlines

Tips for making the best out of your visit

- ✓ Come prepared. Do your research beforehand. Look into professors, labs, and potential research projects available so as to not waste time on those whom you are not interested. Bring along your recommender packet and any other relevant information to use during a meeting.
- ✓ If meeting with particular faculty members are not prearranged, ask the department administrators for help.
- ✓ Talk to the graduate students to find out their perspective on faculty members, program requirements and graduate student life.
- ✓ Be professional. The first impression may be your only impression; be mindful of your behavior, dress, and presentation during the entirety of your visit
- ✓ Take detailed notes!

Preparation for Meeting Potential Advisors

Checklist for Potential Advisor Meetings

1. Literature review on recent papers/publications: research area, significant results, future work, connection to your interests, collaborators, etc
2. Background on the advisor: research interests, previous institutions, possible connections

3. Questions to ask: (you should have at least 2-3 questions you want to ask them)
 - (1) *What open projects do you have? Are these targeted towards a Masters or a PhD? Are they individual or group efforts? What is the funding situation for each project?*
 - (2) *How many students are in the lab? What are your expectations of a graduate student? How often do you meet with students, or have lab meetings?*
 - (3) *Can I take a tour of the lab, meet with some of your graduate students?*
 - (4)
4. Have your 60-second “elevator pitch” ready
 - a. Who are you?
 - b. What is your research/technical background?
 - c. What are your research interests/strengths?
 - d. How can you align yourself with his/her research group?
5. Do not forget to bring your best:
 - CV/Resume
 - Publications
 - Relevant project information
 - Previous presentations
 - Essentially your “recommender packet” and any relevant extras
6. Relax!
7. Make sure to follow up and familiarize yourself with the program’s process, especially if you are interested in a project. Many programs, even at the same institution, handle lab and/or project placement differently.

Tips for Contacting Potential Advisors via Email*

- Know their work before emailing them.
- Write a grammatically-correct, thoughtful, knowledgeable and formal email.
- Refer to a specific paper or work in which you are interested.
- If you do not hear back at first, try again after a week or so. Faculty members are very busy and sometimes miss emails. Do not take it personally.

*These tips apply after offer of admission.

Advisor Selection Tips

Tips for advisor selection: Choosing Dr. Right!

- Remember that he/she will be a key component in the success of your graduate career
- A good advisor/advisee relationship is critical to your personal happiness and professional success
- Consider his/her advising style with how you like to be advised
- Risk v. reward: weigh advising style with research interests; are you willing to sacrifice one for the other?
- Talk to graduate students in the lab; see what the advisor's style is like, the typical program length in their group and where their students end up after the PhD.

Personal Evaluation

Use this personal evaluation to help you start to assess how compatible you may be with your potential advisor. Answer honestly!

1. Do I like to be micromanaged or given the freedom to work independently?
2. Do I work best in the mornings or evenings?
3. Do I work well with people who are very direct (sometimes harsh) or more abstract?
4. Do I work best in a small group/lab versus a large group/lab?
5. Do I need weekly attention from my supervisor or am I okay with once a month?
6. Do I communicate best face to face, via email, telephone, or online?
7. Do I like fast-paced, high-stress environments or low-stress slower environments?
8. Do I like working with younger, new professors or older, tenured professors?
9. Do I enjoy more experimental or theoretical work?
10. Do I mind giving up weekends and/or holidays for working in the lab?
11. Do I work better when the project is already well defined or do I want to define it?
12. Do I like to collaborate across disciplines or am I purely focused on one?
13. Do I value compatibility more than the caliber of research (i.e. am I willing to sacrifice not having one of the above questions affirmed to do research I am interested in?)

Selection Matrix

Personal Rank	Faculty Member <i>-Tenure -Reputation in the field -Management style</i>	Projects of Interest	Lab Impression <i>-Advising Style -Group Dynamics -Group Size -Student Success/Post-Grad Opportunities</i>	School/ Program Impression <i>-Ranking -Location -Quals -Student Life, etc .</i>	Dealbreakers: <i>-Funding? -Quals? -MS or PhD? -Reputation? -Expectations?</i>	Additional Notes:

Personal Rank	Faculty Member <i>-Tenure -Reputation in the field -Management style</i>	Projects of Interest	Lab Impression <i>-Advising Style -Group Dynamics -Group Size -Student Success/Post-Grad Opportunities</i>	School/ Program Impression <i>-Ranking -Location -Quals -Student Life, etc .</i>	Dealbreakers: <i>-Funding? -Quals? -MS or PhD? -Reputation? -Expectations?</i>	Additional Notes:



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